

Advanced Optimization

Chapter 16 — Quadratic Programming

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 16

Quadratic programming: the hinge between theory and the nonlinear solvers

$$\min_x q(x) = \frac{1}{2}x^T Gx + x^T c \quad \text{s.t.} \quad a_i^T x = b_i, \quad i \in \mathcal{E}; \quad a_i^T x \geq b_i, \quad i \in \mathcal{I}. \quad (16.1)$$

- ▶ **Convex QP** ($G \succeq 0$): polynomially solvable, KKT sufficient. **Nonconvex/indefinite QP**: multiple strict local minima, NP-hard in general.
- ▶ Constraints are *linear* $\Rightarrow$ Ch. 12's Lemma 12.7 pays off immediately: **no constraint qualification needed, ever**, in this chapter.

Where QPs come from

- ▶ Directly: Markowitz portfolios ($\max x^T \mu - \kappa x^T Gx$ over a simplex), least squares with linear constraints, model-predictive control — and *DC optimal power flow is exactly a QP*.
- ▶ As an engine: **SQP (Ch. 18) solves one inequality QP per iteration**; interior-point NLP (Ch. 19) solves KKT systems with the same block structure. Whatever we build here runs inside every solver of the last three weeks of this course.

Plan: equality QP and its KKT system (§16.1) $\rightarrow$ three direct solves (§16.2) $\rightarrow$ projected CG (§16.3) $\rightarrow$ inequalities: active-set (§16.5), interior-point (§16.6), gradient projection (§16.7).

Equality QP: one linear system, if the reduced Hessian says yes

$$\min_x \frac{1}{2}x^T Gx + x^T c \quad \text{s.t. } Ax = b \quad (A \in \mathbb{R}^{m \times n}, \text{ rank } m). \quad (16.3)$$

KKT (Theorem 12.1, CQ-free): find (x^*, λ^*) with $\begin{bmatrix} G & -A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}$ (16.4); in step form $x^* = x + p$: $\begin{bmatrix} G & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} -p \\ \lambda^* \end{bmatrix} = \begin{bmatrix} g \\ h \end{bmatrix}$ (16.5), $g = c + Gx$, $h = Ax - b$. Let $Z \in \mathbb{R}^{n \times (n-m)}$ span $\text{Null}(A)$.

Lemma 16.1 + Theorem 16.2

If A has full row rank and the **reduced Hessian** $Z^T GZ$ is positive definite, the KKT matrix K is nonsingular — and the unique (x^*, λ^*) gives the *unique global solution*.

Five lines: $Kw=0$, $w = (w_x, v) \Rightarrow Aw_x = 0 \Rightarrow 0 = w^T Kw = w_x^T Gw_x = u^T Z^T GZ u \Rightarrow u = 0 \Rightarrow A^T v = 0 \Rightarrow v = 0$. Globality: any feasible x has $q(x) = q(x^*) + \frac{1}{2}u^T (Z^T GZ)u > q(x^*)$.

G itself may be indefinite — positive definiteness is demanded only on the null space: Ch. 12's critical cone, now load-bearing. ($Z^T GZ$ merely semidefinite: non-strict minimizer; a negative eigenvalue: (16.4) solvable but the point is a saddle.)

Inertia: how a solver checks second-order conditions without ever forming Z

The KKT matrix is **always indefinite** once $m \geq 1$:

Theorem 16.3

With A of rank m : $\text{inertia}(K) = \text{inertia}(Z^T G Z) + (m, m, 0)$. In particular,
 $Z^T G Z \succ 0 \iff \text{inertia}(K) = (n, m, 0)$.

- ▶ Inertia = (#pos, #neg, #zero eigenvalues); an LBL^T factorization reveals it for free. The working test in every serious solver: factor K , count negatives — exactly $m \Rightarrow$ minimizer step; more $\Rightarrow$ negative curvature in the null space.
- ▶ **File this away for Ch. 19:** IPOPT factors a KKT-like matrix each iteration, reads its inertia, and adds δI to the Hessian block until the inertia is right — the Ch. 3/4/10 “one object, many names” regularization, steered by Theorem 16.3.

Example 16.2 (running example): $n = 3$, $m = 2$, $G = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 5 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $Z = (-1, -1, 1)^T$:

$x^* = (2, -1, 1)^T$, $\lambda^* = (3, -2)^T$, $\text{inertia}(K) = (3, 2, 0)$. ($\lambda_2^* < 0$ is fine — equality multipliers carry no sign; Ch. 12.)

Three direct solves for one KKT system

1. Factor the whole thing: symmetric indefinite LBL^T (16.12)

$P^T K P = L B L^T$, B block-diagonal with $1 \times 1/2 \times 2$ blocks. Cholesky is impossible (indefinite); LU wastes the symmetry — LBL^T is the right tool, about half the cost of sparse LU, and it reports the inertia. Risk: fill-in if the pivoting heuristics fight the sparsity.

2. Schur complement — eliminate x first (needs $G \succ 0$)

$$(AG^{-1}A^T)\lambda^* = AG^{-1}g - h, \quad Gp = A^T\lambda^* - g. \quad (16.13-16.14)$$

Best when G is easy to invert (diagonal/block-diagonal), G^{-1} is on hand (quasi-Newton), or $m \ll n$.

3. Null-space — eliminate the constraints first (needs only Lemma 16.1)

$p = Yp_Y + Zp_Z$, $[Y \mid Z]$ nonsingular:

$$(AY)p_Y = -h, \quad (Z^T G Z)p_Z = -Z^T G Y p_Y - Z^T g, \quad (AY)^T \lambda^* = Y^T (g + Gp). \quad (16.18-16.20)$$

Cholesky on the $(n-m) \times (n-m)$ reduced Hessian — works for indefinite G ; needs a basis Z (QR of A^T , or eliminate variables).

Rule of thumb: $G \succ 0$ and $AG^{-1}A^T$ cheap $\rightarrow$ Schur; factoring G costly but $n-m$ small $\rightarrow$ null-space; large sparse general $\rightarrow LBL^T$ on K .

Iterative solves: Chapter 5 collects another debt

CG on the full system (16.5)? **No** — K is indefinite, CG can break down. (Krylov methods for indefinite systems — GMRES, QMR, LSQR — are legal fallbacks.) The productive route: CG on the *reduced* system,

$$(Z^T GZ) x_Z = -c_Z \quad (16.24)$$

— positive definite, dimension $n - m$: **Algorithm 16.1** is Algorithm 5.3 verbatim, with products $Z^T(G(Zv))$ computed without ever forming $Z^T GZ$, preconditioned by $W_{ZZ} = Z^T HZ$ (16.26).

Algorithm 16.2 (Projected CG): the same iteration without Z

Work in $\mathbb{R}^n$: start from any x with $Ax = b$; replace the preconditioning step by the **projection** $g = Pr$,

$$P = Z(Z^T HZ)^{-1} Z^T, \quad \text{computed as} \quad \begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} g^+ \\ v^+ \end{bmatrix} = \begin{bmatrix} r^+ \\ 0 \end{bmatrix} \quad (16.27, 16.34)$$

(the *constraint preconditioner*, $H = I$: orthogonal projection via $AA^T v = Ar$ (16.31) or the augmented system (16.32)). Every direction stays in $\text{Null}(A)$: **every iterate is feasible**.

No null-space basis, no ill-conditioning from a bad Z ; one factorization involving only A , reused every iteration. This is the engine KNITRO/OOPS expose — the inner solver Ch. 18–19 will call; Steihaug-style negative-curvature exits bolt on when G is indefinite.

Inequality-constrained QP: optimality is settled; the active set is the problem

KKT for (16.1), specialized (16.37): $Gx^* + c = \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* a_i$, feasibility, $\lambda_i^* \geq 0$ on active inequalities. Linear constraints $\Rightarrow$ no LICQ assumption anywhere.

Theorem 16.4 (convex QP: KKT $\Rightarrow$ global)

$G \succeq 0$ and (x^*, λ^*) satisfies (16.37) $\Rightarrow x^*$ is a global solution ($G \succ 0$: unique).

Four lines: for feasible x , $(x - x^*)^T (Gx^* + c) = \sum \lambda_i^* a_i^T (x - x^*) \geq 0$, then $q(x) \geq q(x^*) + \frac{1}{2}(x - x^*)^T G(x - x^*) \geq q(x^*)$.

So what is hard? Knowing $\mathcal{A}(x^*)$. With it, the problem collapses to an equality QP (§16.1–16.3). The three method families are three answers to “how do we find the active set”:

- ▶ active-set: edit a guess, one constraint at a time;
- ▶ interior-point: never commit, follow the central path;
- ▶ gradient projection: let a projected line search rewrite it wholesale.

Two complications

Nonconvex QP (Figure 16.1): several strict local minima, saddles — global solution NP-hard.

Degeneracy (Figure 16.2): dependent active gradients, or weakly active constraints ($\lambda_i^* = 0$) — ambiguity that slows active-set decisions.

The primal active-set method: machinery

Maintain a **working set** $\mathcal{W}_k$: all of $\mathcal{E}$ plus some active inequalities, with *linearly independent* gradients. Iterates stay feasible; q decreases.

Step: equality QP on the working set

With $g_k = Gx_k + c$,

$$p_k = \arg \min_p \frac{1}{2} p^T G p + g_k^T p \quad \text{s.t. } a_i^T p = 0, \quad i \in \mathcal{W}_k \quad (16.39)$$

— solved by any of §16.2–16.3. Constraints in $\mathcal{W}_k$ stay satisfied along p_k for *any* step length.

Step length: walk until something blocks

$$\alpha_k = \min \left(1, \min_{i \notin \mathcal{W}_k, a_i^T p_k < 0} \frac{b_i - a_i^T x_k}{a_i^T p_k} \right) \quad (16.41)$$

If $\alpha_k < 1$, a **blocking constraint** became active — add it to the working set.

Stationarity: read the multipliers

If $p_k = 0$: solve $\sum_{i \in \mathcal{W}_k} a_i \hat{\lambda}_i = g_k$ (16.42). All $\hat{\lambda}_i \geq 0$ on $\mathcal{W}_k \cap \mathcal{I} \Rightarrow$ KKT holds $\Rightarrow$ **global solution** (Theorem 16.4).

Any $\hat{\lambda}_j < 0 \Rightarrow$ drop j (usually the most negative) and continue.

Why dropping a negative multiplier works (Theorems 16.5–16.6)

Theorem 16.5

Let $\hat{x}$ be stationary for working set $\hat{\mathcal{W}}$, gradients independent, $\hat{\lambda}_j < 0$ for some $j \in \hat{\mathcal{W}} \cap \mathcal{I}$. Let p solve the subproblem with j *dropped*. Then $a_j^T p \geq 0$ — the step moves **off** the dropped constraint into feasibility — and under second-order sufficiency $a_j^T p > 0$ strictly, with p a descent direction for q (Theorem 16.6).

Proof in four lines: subtract the two stationarity equations to get $\sum_{i \neq j} (\tilde{\lambda}_i - \hat{\lambda}_i) a_i - \hat{\lambda}_j a_j = Gp$; take inner products with p (which kills every $a_i^T p$, $i \neq j$):

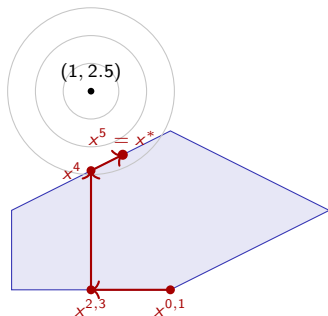
$$-\hat{\lambda}_j a_j^T p = p^T Gp \geq 0 \quad (16.46)$$

and $\hat{\lambda}_j < 0$ forces $a_j^T p \geq 0$. Strict versions follow from $Z^T GZ \succ 0$.

- ▶ This is Ch. 12's sensitivity made operational: **a negative price says the constraint is hurting — release it, and the objective provably improves.**
- ▶ **Finite termination** (strictly convex, nondegenerate): finitely many working sets; each visited at most once since q strictly decreases between stationary points. Cycling is possible under degeneracy in theory, rare in practice; safeguards exist.
- ▶ Cost per iteration: one equality-QP solve — but $\mathcal{W}_k$ changes by *one index*, so factorizations are *updated*, not recomputed (rank-one updates to QR/Cholesky).

Algorithm 16.3, run on a picture (Example 16.4)

$$\min (x_1 - 1)^2 + (x_2 - 2.5)^2 \quad \text{s.t.} \quad \begin{array}{l} (1) \ x_1 - 2x_2 + 2 \geq 0, \quad (2) \ -x_1 - 2x_2 + 6 \geq 0, \\ (3) \ -x_1 + 2x_2 + 2 \geq 0, \quad (4) \ x_1 \geq 0, \quad (5) \ x_2 \geq 0 \end{array}$$



k	$\mathcal{W}_k$	result	action
0	$\{3, 5\}$	$p = 0, \hat{\lambda} = (-2, -1)$	drop 3
1	$\{5\}$	$p = (-1, 0), \alpha = 1$	move
2	$\{5\}$	$p = 0, \hat{\lambda}_5 = -5$	drop 5
3	$\emptyset$	$p = (0, 2.5), \alpha = 0.6$	blocked by 1
4	$\{1\}$	$p = (0.4, 0.2), \alpha = 1$	move
5	$\{1\}$	$p = 0, \hat{\lambda}_1 = 0.8 \geq 0$	stop

Every mechanism appears once: stationarity test, two drops, a free step, a blocking add, and a nonnegative-multiplier certificate at $x^* = (1.4, 1.7)$.

Getting feasible in the first place: Phase-I LP (as in Ch. 13), or a big- M penalty (16.47)/(16.48) — an ℓ_∞/ℓ_1 *exact penalty*, our first sighting of Ch. 17's main tool.

Interior-point for convex QP: Chapter 14's machine with G in the corner

For $\min \frac{1}{2}x^T Gx + c^T x$ s.t. $Ax \geq b$, slacks $y = Ax - b \geq 0$ turn KKT into

$$Gx - A^T \lambda + c = 0, \quad Ax - y - b = 0, \quad y_i \lambda_i = 0, \quad (y, \lambda) \geq 0. \quad (16.55)$$

Perturb complementarity to $Y\Lambda e = \sigma \mu e$, $\mu = y^T \lambda / m$ (**central path**); one Newton step:

$$\begin{bmatrix} G & 0 & -A^T \\ A & -I & 0 \\ 0 & \Lambda & Y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} -r_d \\ -r_p \\ -\Lambda Y e + \sigma \mu e \end{bmatrix}. \quad (16.58)$$

- ▶ **Algorithm 16.4** = Mehrotra predictor–corrector, verbatim from Ch. 14: affine step ($\sigma = 0$) $\rightarrow \sigma = (\mu_{\text{aff}}/\mu)^3 \rightarrow$ corrected step $\rightarrow$ fraction-to-boundary $\tau_k \rightarrow 1$.
- ▶ QP wrinkle: with **unequal** steps, $r_d^+ = (1 - \alpha^{\text{dual}})r_d + (\alpha^{\text{pri}} - \alpha^{\text{dual}})G\Delta x$ (16.65b) can *grow* — the coupling through G . Equal steps are the safe default.
- ▶ Linear algebra: eliminate to an augmented $\begin{bmatrix} * & A^T \\ A & * \end{bmatrix}$ system — LBL^T or projected CG (16.63 $\rightarrow$ Alg 16.2). Same hammers as §16.2–16.3.

Ch. 19 trailer

Everything on this slide survives into nonlinear interior-point — IPOPT is this iteration with G replaced by $\nabla_{xx}^2 \mathcal{L}$, plus inertia correction (Theorem 16.3!) and a globalization. Week 15 will dissect it; today you have seen its heart.

Gradient projection: rewriting the active set wholesale

Active-set methods change $\mathcal{W}_k$ by one index per iteration — 200 active constraints at x^* means ≥ 200 iterations. For **bound constraints** $l \leq x \leq u$ there is a better way.

Stage 1: Cauchy point along the bent path

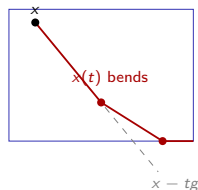
Project the steepest-descent ray onto the box:

$x(t) = P(x - tg, l, u)$ (16.70), piecewise linear with breakpoints $\bar{t}_i$ (16.71). Minimize the piecewise-quadratic $q(x(t))$ interval by interval (coefficients update in $O(1)$ per breakpoint) $\rightarrow$ first local minimizer x^c .

Stage 2: subspace refinement

Fix $x_i = x_i^c$ for $i \in \mathcal{A}(x^c)$; on the free variables (approximately) solve (16.74) — e.g. CG, stopping at the box or projecting back. Only $q(x^+) \leq q(x^c)$ is required for global convergence.

Active set changes by **many indices per iteration**; no convexity needed. The two-stage pattern (Cauchy point + refinement) is Ch. 4 transplanted — it returns in L-BFGS-B and Ch. 18.



The projected path (Figure 16.4): each bend can change several active bounds at once.

Choosing among the three, and the loose ends

	Active-set (§16.5)	Interior-point (§16.6)	Gradient projection (§16.7)
Best size	small/medium	large	large, bounds only
Active-set changes	one per iteration	none (implicit)	many per iteration
Warm starts	excellent	poor–fair	good
Iteration cost	one updated factorization	one KKT factorization	Cauchy + partial CG
Nonconvex G	hard (inertia control)	Ch. 19 territory	natural
Output	exact active set, sharp x^*	interior approach	active set estimate

- ▶ Benchmarks: interior-point wins big problems; **active-set wins sequences of related QPs** — which is exactly what SQP generates. That single row of the table decides Ch. 18's design.
- ▶ Rank-deficient A : detect and prune with SVD / rank-revealing QR of A^T before anything else.
- ▶ Indefinite QP (keywords): negative-curvature directions Zs_Z ride $q \rightarrow -\infty$ until blocked; *inertia-controlling* methods keep ≤ 1 negative eigenvalue in the reduced Hessian at all times.
- ▶ Software: QPOPT/BQPD/QPA (active-set); MOSEK/CPLEX/Gurobi-class and OOQP/QPB (interior-point); KNITRO/OOPS expose projected CG.

Chapter 16 in one table

Object	Statement	Role
KKT system (16.4)	$\begin{bmatrix} G & A^T \\ A & 0 \end{bmatrix}$, indefinite	the one linear system
Lemma 16.1 / Thm 16.2	$Z^T G Z \succ 0 \Rightarrow$ unique global sol.	PD needed only on $\text{Null}(A)$
Thm 16.3 (inertia)	$\text{inertia}(K) = (n, m, 0)$ test	2nd-order check w/o Z ; IPOPT's dial
§16.2 solves	LBL^T / Schur / null-space	pick by structure of G , m vs n
Alg 16.2 projected CG	CG in $\text{Null}(A)$, no Z	Ch. 5 reused; inner engine of 18–19
Thm 16.4	convex: KKT $\Rightarrow$ global	optimality settled; active set is the game
Alg 16.3 active-set	edit $\mathcal{W}_k$ by one; $\hat{\lambda}_j < 0$ steers	finite; warm-startable
Alg 16.4 predictor-corrector	Ch. 14 + G ; equal steps	big QPs; Ch. 19's heart
Alg 16.5 gradient projection	Cauchy point on bent path	bounds; wholesale set changes

Two ideas to carry forward

Positive definiteness is only ever needed on the null space of the constraints — everything in this chapter (Lemma 16.1, inertia, reduced CG) is that one sentence, computed three ways. The inequality-QP problem is really the active-set identification problem; the three method families are three search strategies over subsets.

Homework (1/2)

Derivations

- 1 Prove Theorem 16.2 from (16.4) (fill in the algebra of $q(x) = q(x^*) + \frac{1}{2}p^T Gp$).
- 2 Assemble K for Example 16.2 and verify $\text{inertia}(K) = (3, 2, 0)$ numerically; explain each count via Theorem 16.3.
- 3 Derive the Schur-complement equations (16.13)–(16.14) by block elimination, and state exactly where $G \succ 0$ is used.
- 4 Show that in Algorithm 16.2 all search directions lie in $\text{Null}(A)$ (induction through (16.28)), hence $Ax = b$ throughout.
- 5 Complete the proof of Theorem 16.5: from (16.46), derive $a_j^T p > 0$ under second-order sufficiency, and the descent claim of Theorem 16.6.
- 6 Derive (16.65b) and give a 2-variable example where $\alpha^{\text{pri}} \neq \alpha^{\text{dual}}$ increases $\|r_d\|$.

Homework (2/2)

Implementation

7. Solve Example 16.2 by all three direct methods of §16.2 and by projected CG (Algorithm 16.2, $H = I$); confirm identical (x^*, λ^*) .
8. Implement Algorithm 16.3 with a null-space subproblem solver; reproduce the Example 16.4 trace *exactly* (working sets, steps, multipliers), then re-run from $\mathcal{W}_0 = \{5\}$ and $\mathcal{W}_0 = \emptyset$ and compare paths.
9. Implement Algorithm 16.4; on random convex QPs ($n = 50, 200, 1000$, $m = n/2$), compare iteration counts and wall time against your active-set code — then re-solve a slightly perturbed instance warm-started, and watch the table's warm-start row come true.
10. *Optional, lab-flavored*: write a 3-bus DC-OPF as (16.1), solve with your Algorithm 16.3, and interpret the equality multipliers as locational marginal prices (Ch. 12's (12.80) in the wild).

Next: Ch. 17 — *penalty and augmented Lagrangian methods*.